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(54) **STRAIN RELAXATION LAYER**

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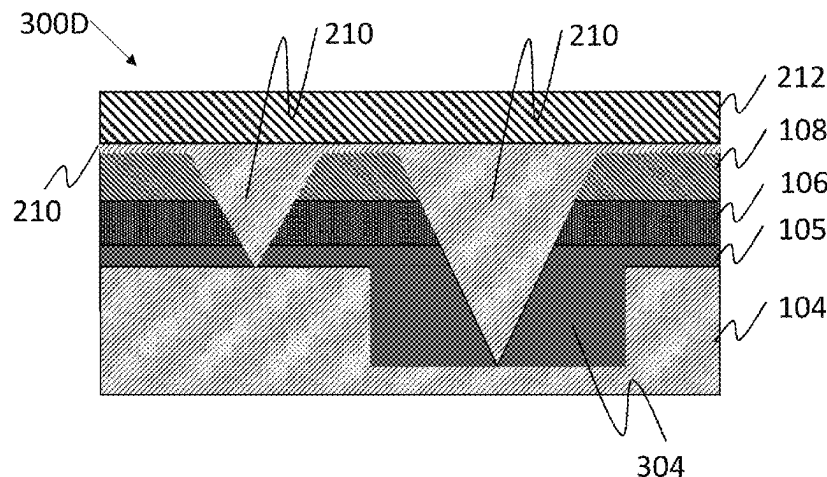
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(57) **ABSTRACT**

A method of forming a strain relaxation layer in an epitaxial crystalline structure, the method comprising: providing a crystalline template layer comprising a material with a first natural relaxed in-plane lattice parameter; forming a first epitaxial crystalline layer on the crystalline template layer, wherein the first epitaxial crystalline layer has an initial electrical conductivity that is higher than the electrical conductivity of the crystalline template layer; forming a second epitaxial crystalline layer on the first epitaxial crystalline layer, wherein the second epitaxial crystalline layer has an electrical conductivity lower than the initial electrical conductivity of the first epitaxial crystalline layer and comprises a material with a second natural relaxed in-plane lattice parameter that is different to the first natural relaxed in-plane lattice parameter of the crystalline template layer;

(Continued)



forming pores in the first epitaxial crystalline layer by electrochemical etching of the first epitaxial crystalline layer to enable strain relaxation in the second epitaxial crystalline layer by plastic deformation of bonds in the first epitaxial crystalline layer and/or at the interface between the first epitaxial crystalline layer and the second epitaxial crystalline layer; and forming one or more channels comprising a conductive material through at least the first epitaxial crystalline layer and the second epitaxial crystalline layer thereby to enable electrical connection to the crystalline template layer through the first epitaxial crystalline layer and the second epitaxial crystalline layer.

20 Claims, 6 Drawing Sheets

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H10H 20/036
See application file for complete search history.

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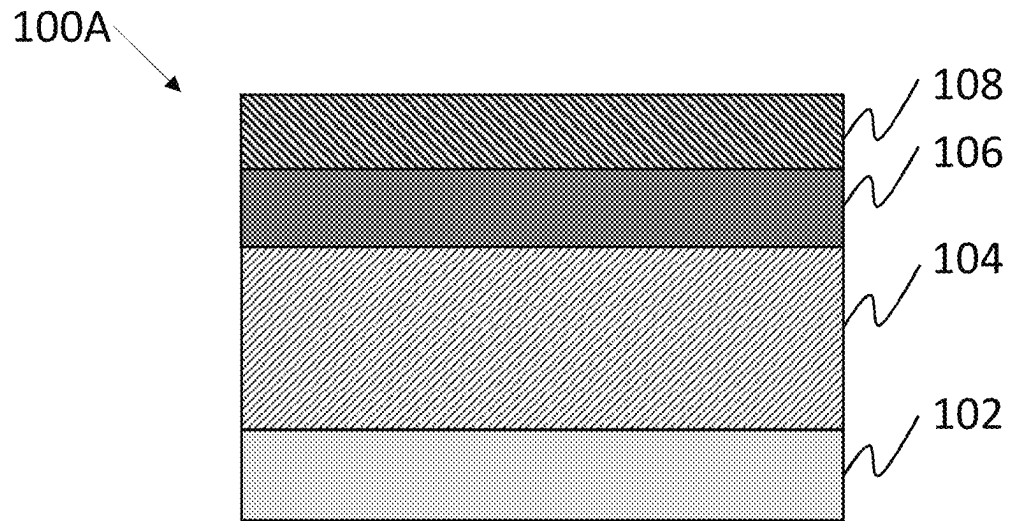


FIGURE 1A

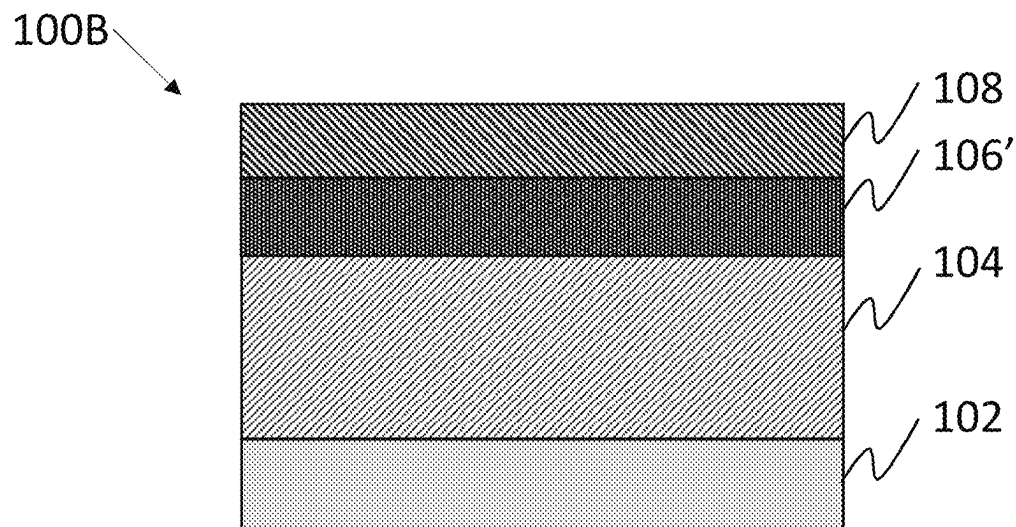


FIGURE 1B

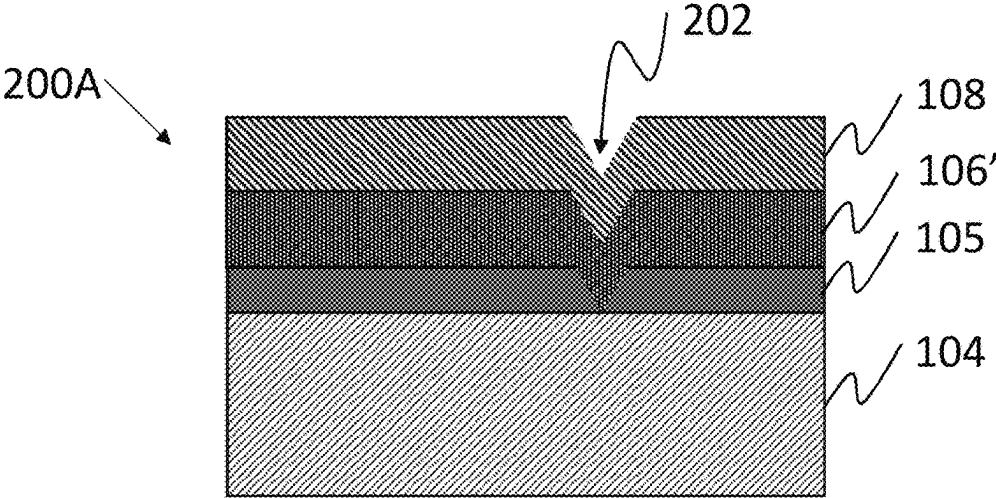


FIGURE 2A

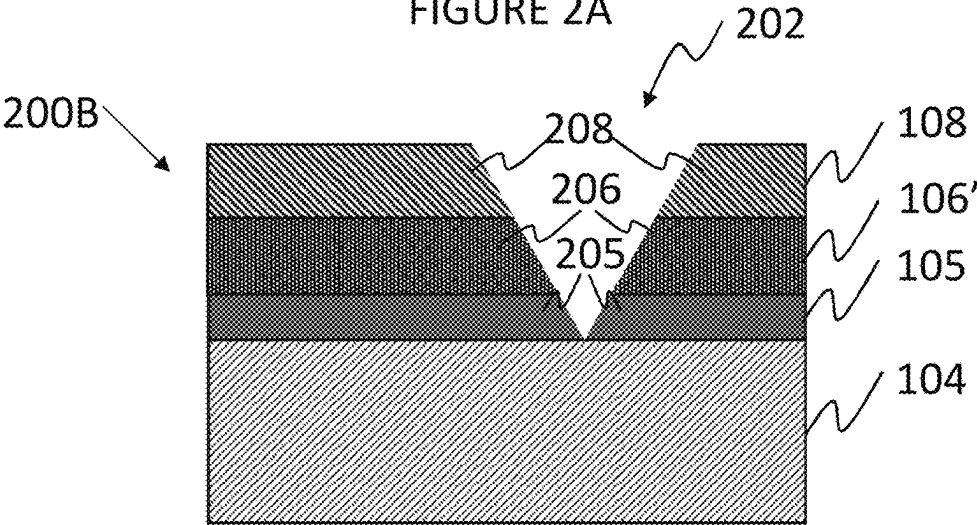


FIGURE 2B

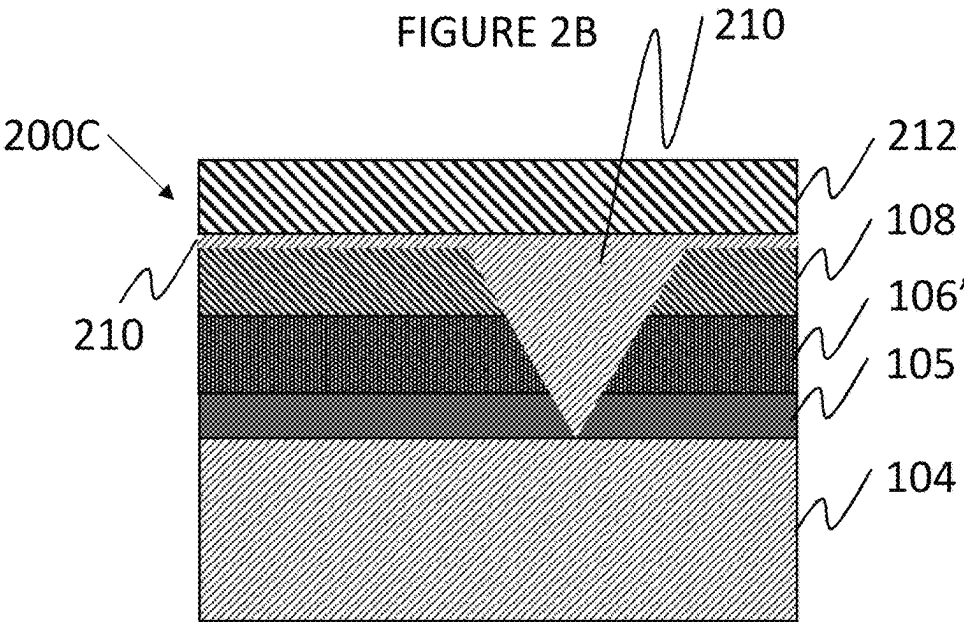
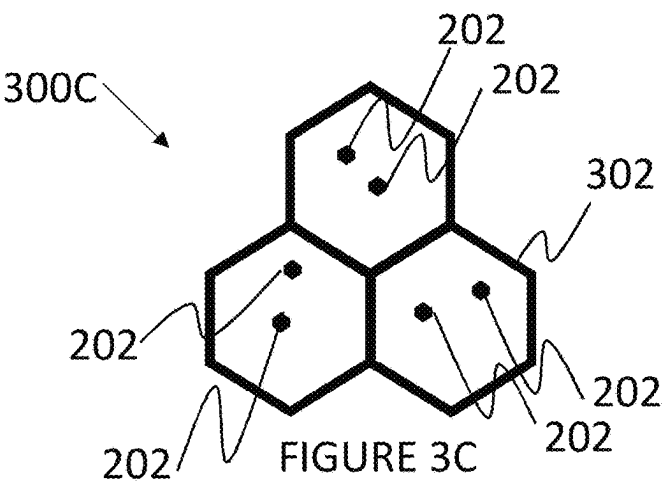
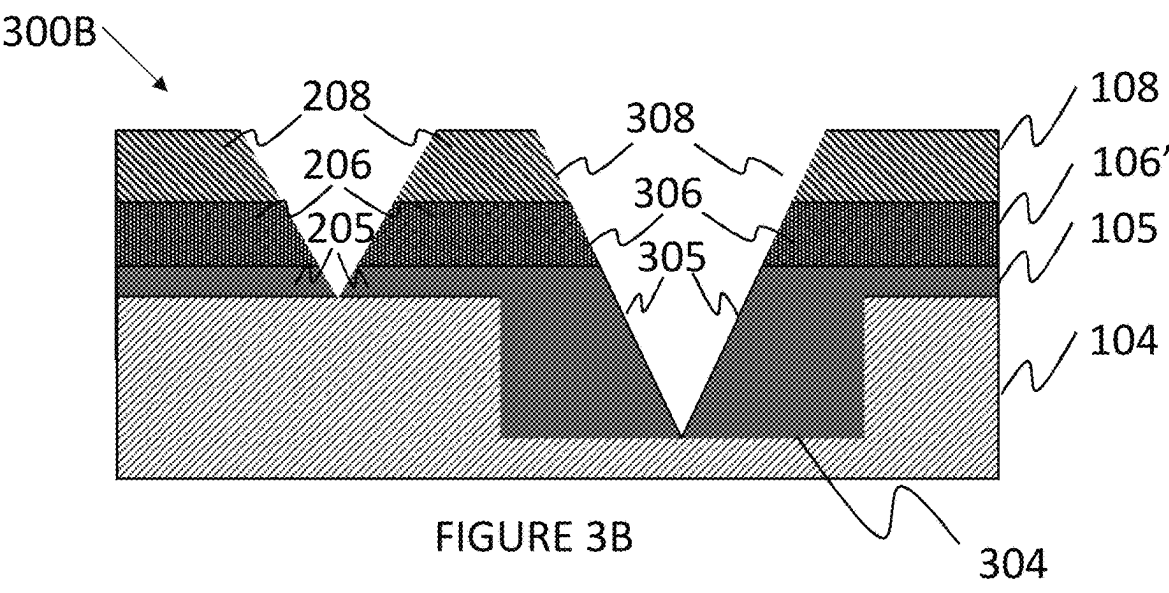
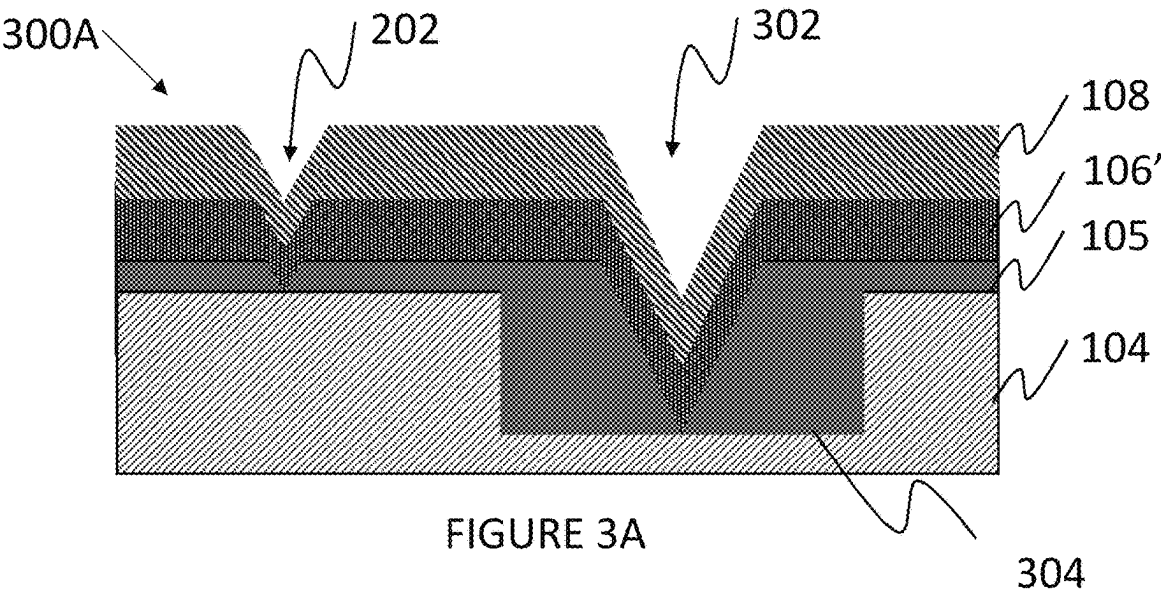
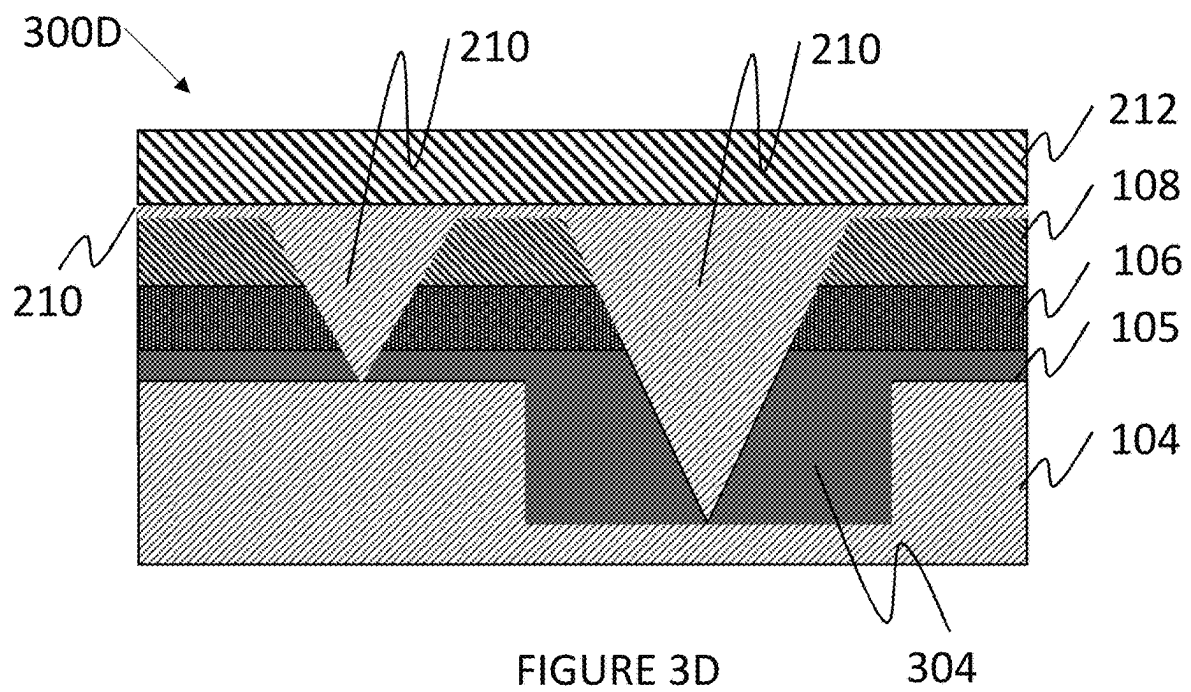


FIGURE 2C





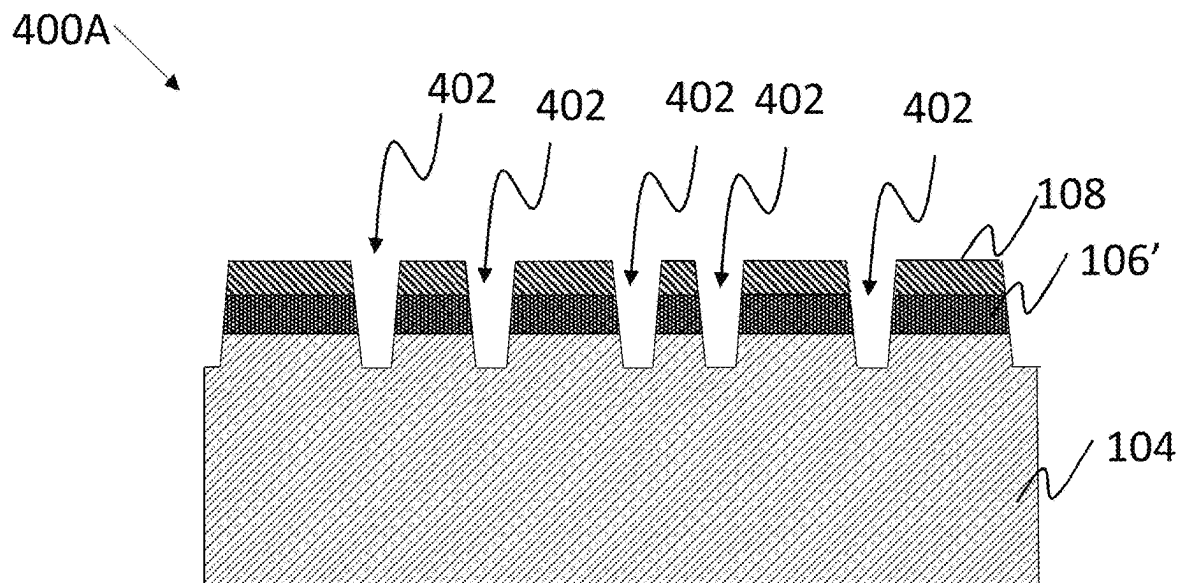


FIGURE 4A

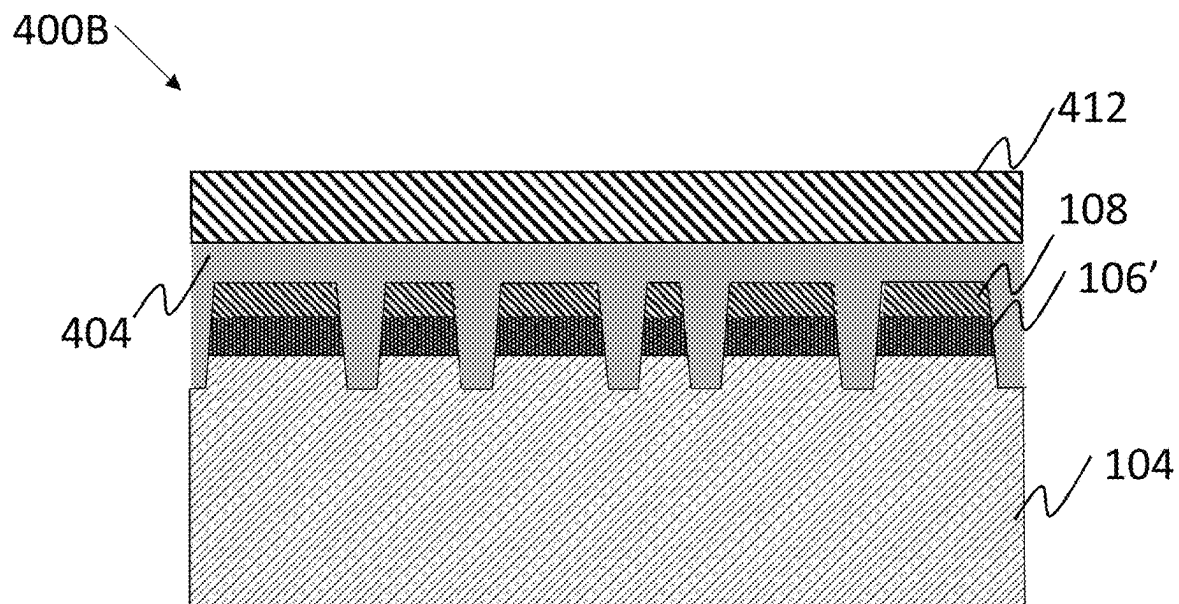
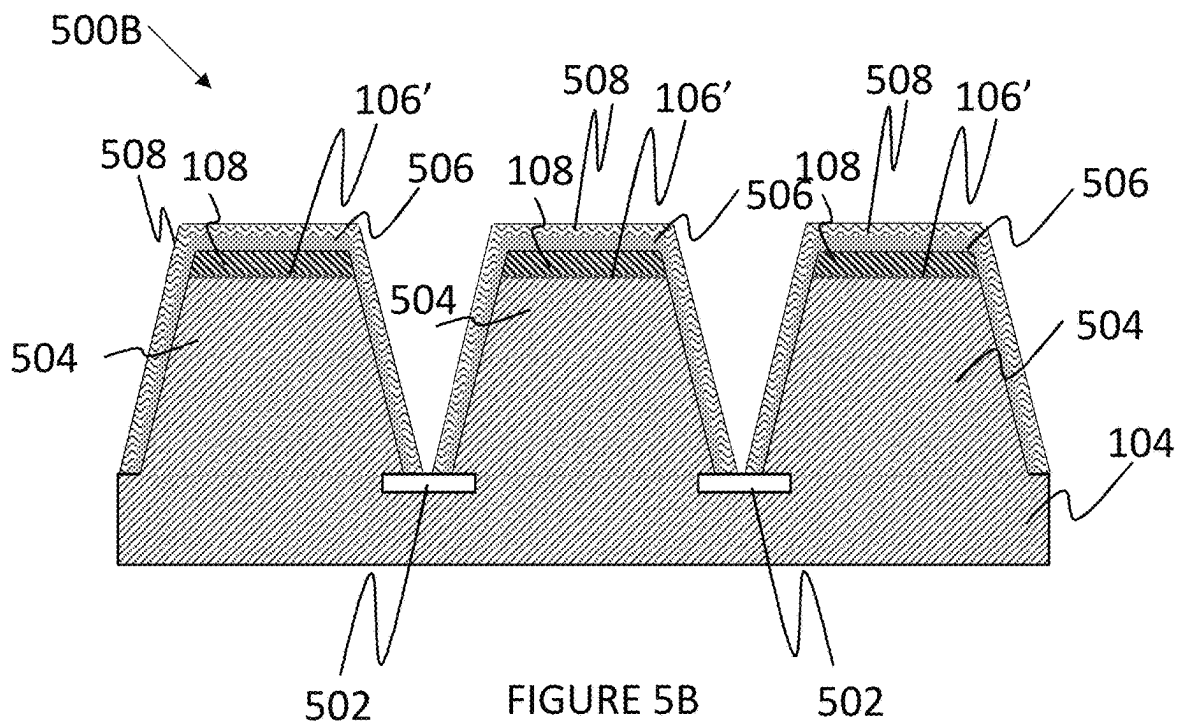
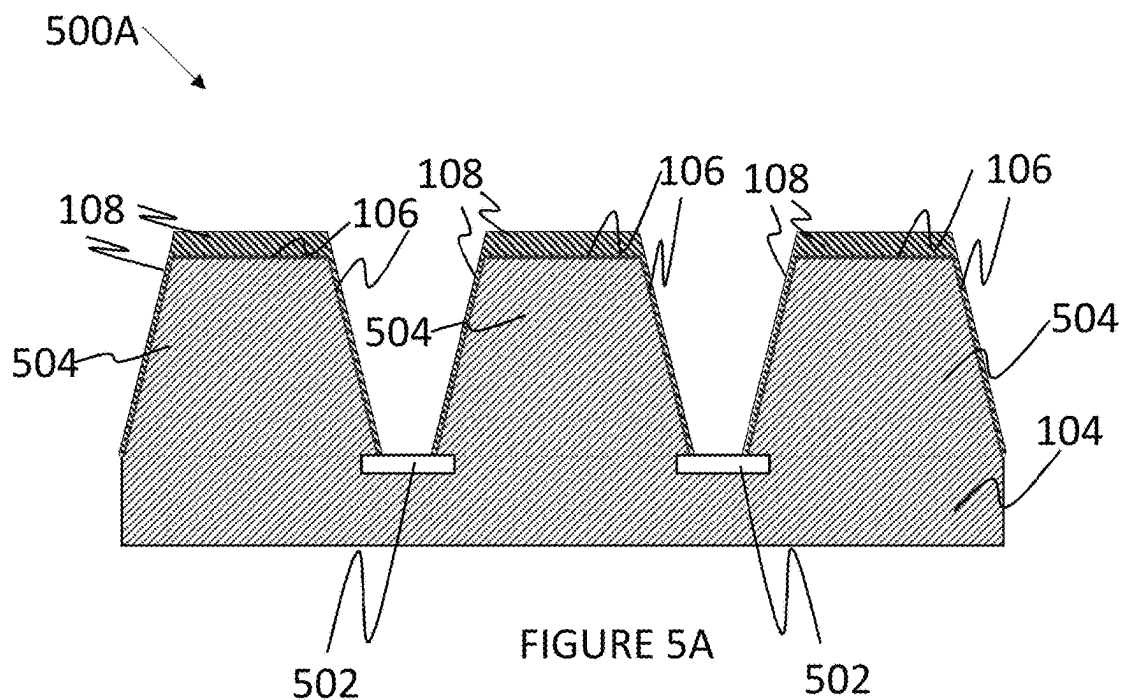


FIGURE 4B



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STRAIN RELAXATION LAYER**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a 35 U.S.C. § 371 national stage application of PCT International Application No. PCT/GB2021/051811, filed on Jul. 14, 2021, which claims the benefit of Great Britain Application No. 2010988.0, filed Jul. 16, 2020, which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

The invention relates to a template and a method for forming a template. In particular, but not exclusively, the invention relates to a template and method of forming a template that enables the growth of strain-relaxed crystalline material for practical device formation.

BACKGROUND OF THE INVENTION

Epitaxially grown crystalline devices, such light emitting diode (LED) devices are typically formed using techniques such as metal organic chemical vapour deposition (MOCVD) and molecular beam epitaxy (MBE) in order to provide crystalline structures with layers of varying composition. The varying composition of the layers, which may be necessitated by the desire to achieve particular emissive recombination, for example, can cause mismatches between the crystalline lattice parameters of the layers. Such mismatches can typically lead to high strain, which can in turn lead to a reduction in internal quantum efficiency (IQE). This is a known problem in the formation of indium gallium nitride (InGaN) based LED structures, particularly in cases where high indium content in InGaN quantum wells (QWs) is used to provide longer wavelength light emission, for example red light emission.

In such cases, attempting to reduce the strain in the underlying layer in order to improve the properties of devices often causes further difficulties. For example, in InGaN based devices, growing QWs on a relaxed InGaN layer with a larger in-plane lattice constant than gallium nitride (GaN), will reduce the strain and therefore improve the IQE. However, achieving high quality relaxed InGaN layers, whilst also enabling the carrier injection in the QWs to form functioning devices, is problematic as high quality n-type layers with the preferred doping levels are typically and advantageously formed from n-type doped GaN and are not easily achieved for materials with the composition needed to reduce strain and therefore improve the IQE of red light emitting InGaN based LEDs.

Porous GaN layers have been proposed as a mechanism to reduce strain in subsequently grown layers, as porous GaN is compliant and rich in broken bonds, which allow the propagation of misfit dislocations at the porous GaN/InGaN interfaces. However, such porous GaN is highly resistive and therefore not generally suitable for the formation of functioning devices.

SUMMARY OF THE INVENTION

In order to mitigate for at least some of the above described problems, there is provided: a method of forming a strain relaxation layer in an epitaxial crystalline structure, the method comprising: providing a crystalline template layer comprising a material with a first natural relaxed

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in-plane lattice parameter; forming a first epitaxial crystalline layer on the crystalline template layer, wherein the first epitaxial crystalline layer has an initial electrical conductivity that is higher than the electrical conductivity of the crystalline template layer; forming a second epitaxial crystalline layer on the first epitaxial crystalline layer, wherein the second epitaxial crystalline layer has an electrical conductivity lower than the initial electrical conductivity of the first epitaxial crystalline layer and comprises a material with a second natural relaxed in-plane lattice parameter that is different to the first natural relaxed in-plane lattice parameter of the crystalline template layer; forming pores in the first epitaxial crystalline layer by electrochemical etching of the first epitaxial crystalline layer to enable strain relaxation in the second epitaxial crystalline layer by plastic deformation of bonds in the first epitaxial crystalline layer and/or at the interface between the first epitaxial crystalline layer and the second epitaxial crystalline layer; and forming one or more channels comprising a conductive material through at least the first epitaxial crystalline layer and the second epitaxial crystalline layer thereby to enable electrical connection to the crystalline template layer through the first epitaxial crystalline layer and the second epitaxial crystalline layer.

There is also provided an epitaxial crystalline structure comprising: a crystalline template layer comprising a material with a first natural relaxed in-plane lattice parameter; a first epitaxial crystalline layer formed on the crystalline template layer; a second epitaxial crystalline layer formed on the first epitaxial crystalline layer, wherein the second epitaxial crystalline layer comprises a material with a second natural relaxed in-plane lattice parameter that is different to the first natural relaxed in-plane lattice parameter of the crystalline template layer; one or more pores formed in the first epitaxial crystalline layer such that any strain in the second epitaxial crystalline layer is relaxed by plastic deformation of bonds in the first epitaxial crystalline layer and/or at the interface between the first epitaxial crystalline layer and the second epitaxial crystalline layer; and one or more channels comprising a conductive material formed through at least the first epitaxial crystalline layer and the second epitaxial crystalline layer thereby to enable electrical connection to the crystalline template layer through the first epitaxial crystalline layer and the second epitaxial crystalline layer.

Advantageously, porosification of an initially high electrical conductivity layer in combination with the formation of one or more channels enables relaxation of a crystalline layer that has a different inherent, relaxed, in-plane lattice parameter compared with the inherent, relaxed, in-plane lattice parameter of an underlying template layer. Such relaxation enables the formation of devices that would otherwise be difficult to achieve, whilst enabling electrical connection with the underlying template layer in a manner that improves device formation.

Preferably, the pores are formed with a density of the volume of the first epitaxial layer between the crystalline template layer and the second epitaxial crystalline layer of greater than 50%, preferably greater than 60% and more preferably greater than 70%. Advantageously, the density of voids in the first epitaxial crystalline layer is controlled in order to facilitate plastic deformation of bonds in the first epitaxial crystalline layer and/or at the interface between the first epitaxial crystalline layer.

Preferably, the second epitaxial crystalline layer comprises at least one V-pit, preferably wherein the depth of the at least one V-pit is substantially the thickness of the first epitaxial crystalline layer and the second epitaxial crystal-

line layer combined. Advantageously, V-pits of such depth enable electrical connection with the underlying crystalline template layer.

Preferably, the method comprises enlarging the at least one V-pit, preferably wherein enlarging the at least one V-pit comprises etching material from the sidewalls of the at least one V-pit thereby to expose at least a portion of the crystalline template layer. Advantageously, V-pits facilitate electrical connection between the crystalline template layer and layers subsequently formed on the second epitaxial crystalline layer through the porosified, electrically highly resistive porous first epitaxial crystalline layer.

Preferably, the method comprises etching through at least the second epitaxial crystalline layer thereby to form one or more islands in the second epitaxial crystalline layer. Advantageously, the formation of islands controls relaxation of the second epitaxial crystalline layer.

Preferably, the method comprises patterning the second epitaxial crystalline layer, wherein patterning comprises at least one of a lithographic technique and a self-assembled Ni hard mask. Advantageously, patterning the second epitaxial crystalline layer with a self-assembled Ni hard mask enables the formation of narrow channels for strain relaxation and electrical connection with the underlying crystalline template layer.

Preferably, the method comprises forming the crystalline template layer on a patterned template layer. Advantageously, by controlling the underlying morphology of the crystalline template layer, improved device formation is facilitated. Beneficially, the formation of LED devices and arrays of LED devices, such as high resolution micro LED arrays with improved light strain management, more efficient light generation and more efficient light extraction can be provided.

Preferably, the patterned template layer is patterned to provide trenches, preferably wherein forming the crystalline template layer comprises forming overgrowth material in the trenches to provide v-grooves, thereby to enable the formation of two dimensional domains at least partially defined by valleys in the second epitaxial crystalline layer. Advantageously, the formation of trenches enables the size of valleys to be tuned by overgrowth of material in the trenches.

Preferably, the patterned template layer is patterned to provide one or more overgrowth structures, preferably wherein forming the patterned crystalline template layer comprises at least partially masking the crystalline template layer to provide one or more apertures and forming overgrowth material in the one or more apertures thereby to provide the one or more overgrowth structures. Advantageously, overgrowth on a patterned substrate can be used to provide structures protruding from an initially substantially flat planar substrate. Such morphology can be used to enable improved light emission properties from LED devices in combination with improved strain management of the underlying crystalline material upon which the LED devices are formed.

Preferably, where the patterned template layer is patterned to provide one or more overgrowth structures, forming the one or more channels comprising the conductive material through at least the first epitaxial crystalline layer and the second epitaxial crystalline layer comprises removing material from one or more sidewalls of the one or more overgrowth structures. Advantageously, removal of material to form channels through at least the first epitaxial crystalline layer and the second epitaxial crystalline layers enables

electrical connection to the crystalline template layer through the first epitaxial crystalline layer and the second epitaxial crystalline layer.

Preferably, the method comprises planarizing at least a portion of the second epitaxial crystalline layer thereby to form a third epitaxial crystalline layer that is pseudomorphic with the second epitaxial crystalline layer. Advantageously, the third epitaxial crystalline layer provides a surface for the formation of functioning devices.

Preferably, the method comprises forming material in the one or more channels, preferably wherein the material forms part of the third epitaxial crystalline layer. Preferably, the third epitaxial crystalline layer is an electrically conductive layer, preferably the third epitaxial crystalline layer is in electrical communication with the crystalline template layer and/or wherein the third epitaxial layer forms part of an active region configured to emit light in response to carrier recombination. Advantageously, devices formed on the third epitaxial crystalline layer can be contacted and in electrical connection with the underlying crystalline template layer. Beneficially, where the third epitaxial layer forms part of an active region configured to emit light in response to carrier recombination, the efficiency of light generation is improved. Advantageously, where the third epitaxial layer forms part of an active region on the sidewalls of features of the patterned crystalline template layer, improved electrical connection and carrier injection into the third epitaxial crystalline layer is provided which coupled with the strain relaxed second epitaxial crystalline layer provides for improved light generation from devices formed at least in part on the second epitaxial crystalline layer.

Preferably, at least one of the crystalline template layer, the first epitaxial crystalline layer, the second epitaxial crystalline layer and the third epitaxial crystalline layer comprises a semiconductor material, preferably wherein the semiconductor material is a III-V based material, more preferably wherein the III-V material is a nitride based material. Advantageously, techniques for the provision of such materials can provide high quality crystalline material with low defect density.

Preferably, the epitaxial crystalline structure forms part of a light emitting device. Advantageously, strain relaxation is managed in combination with electrical conductivity to form high quality practical devices.

Preferably, the light emitting device forms part of an array of light emitting pixels. Advantageously, arrays of light emitting pixels can be formed on the strain relaxed layer such that the in-plane lattice parameter is suited to the material requirements hence providing higher quality devices that can be electrically connected on both sides, thereby increasing the density of pixels in the array.

Further aspects of the invention will be apparent from the description and the appended claims.

DETAILED DESCRIPTION OF AN EMBODIMENT OF THE INVENTION

A detailed description of embodiments of the invention is described, by way of example only, with reference to the Figures, in which:

FIG. 1A shows a cross sectional view of an epitaxial crystalline structure;

FIG. 1B shows a cross sectional view of the epitaxial crystalline structure of FIG. 1A that has been processed further;

FIG. 2A shows a cross sectional view of an epitaxial crystalline structure comprising V-pits;

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FIG. 2B shows a cross sectional view of the epitaxial crystalline structure of FIG. 2A that has been processed further;

FIG. 2C shows a cross sectional view of the structure of FIG. 2B that has been processed further;

FIG. 3A shows a cross sectional view of an epitaxial structure comprising V-pits and V-grooves;

FIG. 3B shows a cross sectional view of the structure of FIG. 3A that has been processed further;

FIG. 3C shows a plan view of the structure of FIG. 3B;

FIG. 3D shows a cross sectional view of the structure of FIG. 3B that has been processed further;

FIG. 4A shows a cross sectional view of the structure of FIG. 1B that has been processed further;

FIG. 4B shows a cross sectional view of the structure of FIG. 4A that has been processed further;

FIG. 5A shows a cross sectional view of an epitaxial structure formed on a template; and

FIG. 5B shows a cross sectional view of the structure of FIG. 5A that has been processed further.

The formation of strain relaxed epitaxial crystalline layers on which functioning solid state devices can be formed is described. The structure, and the method for forming the structure, address at least some of the problems described above. FIGS. 1 to 5 describe the formation of strain relaxed layers in epitaxial crystalline structures where the strain relaxed layers have an inherent, natural lattice parameter that is different from the inherent, natural, lattice parameter of the substrate upon which the strain relaxed layer is formed. The formation of such strain relaxed layers means that subsequently formed layers with an inherent crystalline lattice parameter more closely aligned to the strain relaxed layers can be formed and will therefore be able to be formed with higher crystalline quality and reduced defects.

The method and structure described herein are based on III-V crystalline materials, in particular nitride based semiconductor materials. However, in further examples, the skilled person understands that the techniques described here can be applied to different crystalline structures and semiconductor materials, such as other III-V crystalline materials, or II-VI crystalline materials. Advantageously, not only does the structure provide a template layer for forming materials with natural lattice parameters more closely aligned to the template than the underlying bulk substrate layers, but it also allows for the materials with natural lattice parameters more closely aligned to the template to be formed on an electrically conductive template, therefore enabling practical functioning devices of high crystalline quality to be formed, for example high resolution arrays of light emitting diode (LED) devices.

FIG. 1A shows a cross sectional view of an epitaxial structure 100A. The epitaxial structure 100A provides an initial template for forming devices and is formed by metal organic chemical vapour deposition (MOCVD). In further examples, alternative and/or additional growth and/or deposition techniques are used to provide the epitaxial layers described herein. In an example, molecular beam epitaxy (MBE) is used. Light emitting diode (LED) devices are typically formed by providing a growth substrate, upon which multiple epitaxial layers of semiconductor crystalline material are grown and/or deposited to form functioning devices. For gallium nitride (GaN) based LED devices, n-type doped n-GaN is typically provided as a base layer for the formation of p-n junctions. The epitaxial structure 100A provides a structure that is processed in order to enable the formation of devices, such as LED devices with high quality

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crystalline material, whilst enabling the devices to be formed on an electrical conductive layer.

In FIG. 1A, there is shown a growth substrate 102. The growth substrate 102 is formed from silicon. In further examples, alternative and/or additional materials are used to form the substrate, such as sapphire, silicon carbide, or any other suitable substrate material. Upon the growth substrate 102, n-type doped n-GaN is provided and effectively acts as a crystalline template layer 104. Whilst the crystalline template layer 104 is shown as a n-type GaN layer, in further examples, alternative and/or additional layers, such as buffer layers, are included in order to control the properties of the n-GaN layer (such as crystalline quality, doping level, thickness etc.). The crystalline template layer 104 is approximately 1000 nm thick and is doped at a concentration of 5×10^{18} at/cm³. In further examples, the crystalline template layer 104 is formed to a different thickness and has a different doping concentration whilst still enabling the functionality described herein. In further examples, the crystalline template layer 104 is a p-type doped layer formed in order to provide the functionality described herein and the composition of the subsequently formed layers is adjusted accordingly.

On top of the n-type crystalline template layer 104 there is formed a first epitaxial crystalline layer 106, the first epitaxial crystalline layer 106 is formed from gallium nitride (GaN). Accordingly, the first epitaxial crystalline layer 106 has the same natural, relaxed, in-plane lattice constant as the crystalline template layer 104. In further examples, the first epitaxial crystalline layer 106 is formed from a different material and is pseudomorphic with the crystalline template layer 104. The first epitaxial crystalline layer 106 is formed as a highly doped layer, where the doping is such that there is an initial doping contrast between the highly doped first epitaxial crystalline layer 106 and the n-type crystalline template layer 104. Such contrast allows for the porosification of the first epitaxial crystalline layer 106 in a subsequent step, describe below with respect to FIG. 1B. The first epitaxial crystalline layer has a thickness of approximately 30 nm and is initially doped at a concentration of the order of 1×10^{20} at/cm³. Such a doping concentration is achieved using Si. In further examples, co-doping using Si and Al is used to achieve such a doping concentration, while preventing excessive surface roughening. In further examples, different techniques, concentrations and thicknesses are alternatively and/or additionally used to form the first epitaxial crystalline layer 106. Advantageously, the use of a thin first epitaxial crystalline layer 106 of the order of 30 nm provides a highly defective layer after the porosification treatment, where broken bonds allow for movement of misfit dislocations. In further examples, the first epitaxial crystalline layer 106 has a thickness of between 1 nm and 300 nm, preferably between 20 nm and 50 nm. In further examples, the first epitaxial crystalline layer 106 is initially doped with Si at a concentration of between 8×10^{18} at/cm³ and 5×10^{20} at/cm³ and preferably between 3×10^{19} at/cm³ and 1×10^{20} at/cm³. In further examples, the first epitaxial crystalline layer 106 is co-doped with Al and Si with a mole fraction between 0.5% to 10%, and preferably between 1% and 3%.

Upon the first epitaxial crystalline layer 106 there is provided a second epitaxial crystalline layer that is an undoped layer 108, the undoped second epitaxial crystalline layer 108 is a layer with an inherently different lattice constant to the crystalline template layer 104 and the first epitaxial crystalline layer 106. In the example of FIG. 1A, the second epitaxial crystalline layer 108 is an undoped layer of indium gallium nitride (InGaN). The undoped second

epitaxial crystalline layer **108** is approximately 100 nm to 150 nm thick. Advantageously, the second epitaxial crystalline layer **108** has a thickness that enables elastic relaxation. Accordingly, the growth of the second epitaxial crystalline layer **108** results in a second epitaxial crystalline layer **108** that is initially strained by the underlying material of the crystalline template layer **104** and the first epitaxial crystalline layer **106** such that the different inherent, natural, in-plane lattice parameter of the second epitaxial crystalline layer **108** is pseudomorphically aligned with the inherent, natural, in-plane lattice parameter of the first epitaxial crystalline layer **106**. In further examples, the second epitaxial crystalline layer **108** is formed with a different thickness, whilst enabling the functionality described herein. In an example, the second epitaxial crystalline layer **108** has a thickness between 100 nm and 500 nm. In further examples, the second epitaxial crystalline layer **108** has a thickness preferably between 150 nm and 200 nm. The second epitaxial crystalline layer **108** is a bulk layer of InGa_N with In composition of between 0% and 25% and preferably between 6% and 15%. Whilst the second epitaxial crystalline layer **108** is a bulk InGa_N layer, in further examples, the second epitaxial crystalline layer **108** is a superlattice structure of multiple layers with alternating chemical composition that provides an average indium composition in InGa_N of between 0% and 25% and preferably between 6% and 15%. In an example, when the InGa_N second epitaxial crystalline layer is a superlattice structure, the inherent different lattice constant to the crystalline template layer **104** is related to the average composition over the whole superlattice layer forming the second epitaxial crystalline layer.

The provision of such an epitaxial structure **100A** with a high doping contrast between the first epitaxial crystalline layer **106**, the crystalline template layer **104** and the second epitaxial crystalline layer **108** enables porosification of the initially highly doped first epitaxial crystalline layer **106**. The porosification of the first epitaxial crystalline layer **106** occurs by an electrochemical process. Electrochemical etching processes can be used selectively to etch epitaxial layers such that the most conductive layer is etched first. The porosification of buried layers in an epitaxial structure, such as the epitaxial structure **100A** of FIG. 1A can take place through threading dislocation cores in the second epitaxial crystalline layer **108**. Advantageously, the topmost second epitaxial crystalline layer **108** need not be attacked and there is no need to pattern the second epitaxial crystalline layer **108** in order to porosify subsurface layers. Whilst the second epitaxial crystalline layer **108** is described as an undoped layer **108**, in further examples, the undoped second epitaxial crystalline layer **108** has a level of doping that provides sufficient contrast in conductivity to enable the porosification of a subsurface layer in accordance with the methods described herein.

Electrochemical etching of the epitaxial structure **100A** results in porosification of the first epitaxial crystalline layer **106** through threading dislocations or other openings in the undoped InGa_N of the second epitaxial crystalline layer **108** enabling such electrochemical processing of the subsurface first epitaxial crystalline layer **106** to take place. This is facilitated by the high doping contrast between the n-type crystalline template layer **104** and first epitaxial crystalline layer **106** meaning that only the first epitaxial crystalline layer **106** becomes porosified during an electrochemical etching of this layer. Advantageously, the doping contrast enables control of the porosification process since, as the first epitaxial crystalline layer **106** has pores formed in it, the porosified first epitaxial crystalline layer **106'** becomes

highly resistive and therefore the electrochemical process used to porosify the first epitaxial crystalline layer **106** stops forming pores.

Accordingly, once the epitaxial structure **100A** has been processed by electrochemical etching in order to provide a porous first epitaxial crystalline layer **106'**, the first epitaxial crystalline layer **106** becomes a porous and compliant epitaxial crystalline layer **106'**.

FIG. 1B shows a processed version of epitaxial structure **100A**. Epitaxial structure **100B** is a processed version of the epitaxial structure **100A** whereby the first epitaxial crystalline layer **106** has been porosified in order to provide pores throughout the porous first epitaxial crystalline layer **106'**. Such porosification enables strain relaxation of the second epitaxial crystalline layer **108** of undoped InGa_N. Such strain relaxation requires the second epitaxial crystalline layer **108** to expand. The lateral expansion of the second epitaxial crystalline layer **108** is controlled in order to provide advantageous structures, as described below. The extent to which the porosified first epitaxial crystalline layer **106'** is, by density of volume, pore and Ga_N material, depends on the electrochemical etching process. The electrochemical etching process, whilst beneficially providing compliant material in a subsurface layer, results in the porosified first epitaxial crystalline layer **106'** having a reduced electrical conductivity compared with the highly doped conductivity of the first epitaxial crystalline layer **106** when initially formed.

As described above, the second epitaxial crystalline layer **108**, which is initially strained to the first epitaxial crystalline layer **106**, should expand in order to relax. The expansion of the second epitaxial crystalline layer **108** is controlled by the formation of appropriate gaps and/or channels, as described with reference to FIGS. 2 to 5.

At FIG. 2A there is shown a cross sectional view of an epitaxial structure **200A** that is provided in accordance with processed epitaxial structure **100B** described with respect to FIG. 1B, such that there is a crystalline template layer **104**, a porosified first epitaxial crystalline layer **106'** and second epitaxial crystalline layer **108**. The epitaxial structure **200A** further shows a V-pit **202** in the surface of the second epitaxial crystalline layer **108**. Such V-pit **202** is created in the formation of the n-type crystalline template layer **104**. FIG. 2A shows a top layer **105** of the crystalline template layer **104**. The top layer **105** is also formed from Ga_N, however, the formation of V-pits in the top layer **105** is enabled by changing the growth conditions when forming the crystalline template layer **104**. For example, V-pits in the top layer **105** of the crystalline template layer **104** can be initiated by lowering the growth temperature. Such a top layer **105** is of the order of 50 nm thick. Advantageously, such a thickness generates V-pits in subsequent layers of an appropriate thickness to enable channels to be formed through the epitaxial structure **200A**. In further examples, the top layer **105** has a configuration designed to provide the functionality described herein. The cross sectional view of the epitaxial structure **200A** shows one V-pit, however, in further examples, additional V-pits are formed across the planar surface of the top layer **105** of the crystalline template layer. In an example, such V-pits are formed randomly across the top layer **105** of the crystalline template layer.

The formation of V-pits in the top layer **105** of the crystalline template layer **104** results in V-pits that propagate through subsequently grown layers. Accordingly, V-pit **202** is formed in the surface of the second epitaxial crystalline layer **108** in accordance with the V-pit formed in the top layer **105** of the crystalline template layer **104**. In further

examples, additional and/or alternative techniques are used to form V-pits **202** in the second epitaxial crystalline layer **108**.

Once one or more V-pits **202** are formed in the second epitaxial crystalline layer **108**, the epitaxial structure **200A** is processed in order to enlarge the V-pits **202**, thereby to open one or more channels and expose at least part of the crystalline template layer **104** and/or the top layer **105** of the crystalline template layer **104**. This is shown at FIG. **2B**, where the sidewalls of the V-pits **202** are shown to have been etched.

FIG. **2B** shows a cross sectional view of an epitaxial structure **200B**, which is the epitaxial structure **200A** of FIG. **2A** that is been processed further. There is shown at FIG. **2B** the enlargement of the V-pit **202** by etching of the sidewalls of the V-pit **202**. Processing of the epitaxial structure **200A** of FIG. **2A** by wet etching enables anisotropic etching of the epitaxial structure **200A**. Wet etching is performed using potassium hydroxide (KOH) at an elevated temperature. In further examples, additional and/or alternative techniques are used in order to etch and enlarge the V-pits **202**. For example, tetramethylammonium hydroxide (TMAH) is used.

As shown at FIG. **2B**, the sidewalls **208** of the second epitaxial crystalline layer **108** of undoped InGaN are etched to provide an open path to the conductive GaN. Subsequently the sidewalls of **206** of the porosified first epitaxial crystalline layer **106** and the sidewalls **205** of the V-pitted top layer **105** are enlarged. The resultant etched V-pit **202** is substantially the depth of the thickness of the porosified first epitaxial crystalline layer **106'** combined with the thickness of the second epitaxial crystalline layer **108** of undoped InGaN. Advantageously, such enlargement provides a route for electrical connection to the n-type crystalline template layer **104** through the epitaxial structure **200B**. Once the sidewalls **205**, **206**, **208** of the V-pit **202** have been etched to provide one or more channels through at least the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108**, the epitaxial structure **200B** is processed further, as shown at FIG. **2C**.

FIG. **2C** shows a cross sectional view of an epitaxial structure **200C**, which is the epitaxial structure **200B** of FIG. **2B** that has been processed further. At FIG. **2C** there is shown a conductive material that is an n-type material **210** formed in the channel provided by the enlarged V-pit **202** described with reference to FIGS. **2A** and **2D**. The n-type material **210** also forms a third epitaxial crystalline layer that is a thin pseudomorphic layer of n-type material **210** on top of the second epitaxial crystalline layer **108** of undoped InGaN. The n-type material **210** is in electrical contact with the n-type material of the crystalline template layer **104**, beneficially enabling current spreading and the facility to make contact via the n-doped side of the epitaxial structure **200C** in subsequently formed devices. The n-type material **210** planarizes the structure. When formed of GaN, the planarized layer of n-type material **210** is under tensile stress when the second epitaxial crystalline layer **108** of undoped InGaN is relaxed. Whilst the conductive n-type material **210** is formed of GaN, in further examples, the conductive material **210** is formed from a different material. In an example, the conductive n-type material **210** is formed from InGaN with an In composition of between 0% and 6%. The conductive material **210** is formed as a layer with a thickness of between 5 nm and 300 nm and preferably between 20 nm and 100 nm.

Whilst the conductive material formed in the one or more channels provided by the V-pits **202** is an n-type material **210** that is shown to planarize the structure, in further

examples, alternatively or additionally, the conductive n-type material **210** partially planarizes the second epitaxial crystalline layer **108** or does not planarize the second epitaxial crystalline layer **108**, whilst still at least partially filling the one or more channels provided by V-pits **202** with conductive material thereby to provide conductivity to relatively strain relaxed regions formed subsequently to the relaxation of the second epitaxial crystalline layer **108**.

On top of the n-type material **210** there is formed a further layer **212**. The further layer **212** is, in an example, an epitaxial structure of multiple layers with an active region associated with the provision of red light from InGaN based quantum wells (QWs). Advantageously, the further layer **212** benefits from being formed on an electrically conductive layer that is pseudomorphic with the second epitaxial crystalline layer **108**. Whilst the further layer **212** is shown to be provided on the n-type material **210**, in further examples the further layer **212** is provided on both the n-type material **210** and the second epitaxial crystalline layer **108**. In further examples, the further layer **212** comprises one or more LED devices, which, in a further example, form an array of devices, such as a high resolution array of microLED devices. In further examples, the further layer **212** alternatively and/or additionally forms part of the n-type material **210** and/or replaces the n-type material, thereby beneficially to enable carrier injection directly into the portion of the strain relaxed material formed on the second epitaxial crystalline layer **108** through the further layer **212** formed at least partially on a sidewall of the n-type crystalline template layer **104**, such as the V-pitted top layer **105** of the crystalline template layer **104**.

Whilst the enlarged V-pit **202** of the structures **200** is described with respect to FIG. **2A** to **2C** to provide routes to enable conductive material to be formed on the second epitaxial crystalline layer **108**, and facilitate strain relaxation of the second epitaxial crystalline layer **108**, the location and density of V-pits **202** in top layer **105** of the crystalline template layer **104** are, to an extent, random. The density of V-pits **202** in the top layer **105** is controlled using appropriate growth techniques to enable electrical connection in a controlled manner.

FIGS. **3A** to **3D** show further structures that are incorporated into the epitaxial structure **200A** to **C** of FIGS. **2A** to **2C** to provide further control of the resultant template structure.

FIG. **3A** shows a cross sectional view of an epitaxial structure **300A**. The epitaxial structure **300A** is the structure **200A** of FIG. **2A** that has been processed using an additional step.

FIG. **3A** shows a trench **304** formed in the crystalline template layer **104**. The trench **304** is formed using a lithographic technique that uses patterning and etching steps to form the trench **304**. In further examples, the crystalline template layer **104** is processed using different techniques to provide the trench **304**. The trench **304** is formed in accordance with any beneficial structure in the crystalline template layer **104** that is to be subsequently processed.

Once a trench **304** is formed in the crystalline template layer **104**, the subsequent layers **105**, **106**, **108** are epitaxially formed. Such formation results in a V-groove **302** in the epitaxial structure **300A** that corresponds to the trench **304**. V-pits are additionally formed in the top layer **105** of the crystalline template layer **104** and propagate through the layers to provide the V-pit **202** in addition to the V-groove **302**. In further examples, V-grooves **302** are provided without V-pits **202**.

Whilst the trench **304** is shown to be formed in the crystalline template layer **104** prior to forming the subsequent epitaxial layers, thereby to enable the formation of channels through at least the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108**, in further examples the trench **304** is formed by patterning the crystalline template layer **104** with a mask and forming overgrowth material on the mask-patterned crystalline template layer **104**. Such a technique is used to provide an array of columns each having a regular trapezoidal cross section normal to the substrate, for example as described in WO 2020/008200 A1. Overgrowth on such a mask-patterned crystalline template layer **104** is used in further examples to provide underlying layers for the formation of an array of light emitting devices, as described in further detail with respect to FIGS. **5A** and **5B**. In further examples, the morphology of the crystalline template layer **104** is controlled by patterning and/or other techniques to provide appropriate underlying substrate for improved device formation.

Once V-grooves **302** have been formed, the process moves on analogously to that shown with respect to FIG. **2**. At FIG. **3B** there is shown a cross sectional view of an epitaxial structure **300B** that is the epitaxial structure **300A** of FIG. **3A** that has been processed further.

There is shown at FIG. **3B** the enlargement of the V-pit **202** by etching of the sidewalls and the enlargement of the V-groove **302** by etching of the sidewalls. Processing of the epitaxial structure **300A** of FIG. **3A** by wet etching enables anisotropic etching of the epitaxial structure **300A**. As shown at FIG. **3B**, the sidewalls **308** of the second epitaxial crystalline layer **108** are etched. Subsequently the sidewalls of **306** of the porosified first epitaxial crystalline layer **106'** and the sidewalls **305** of the V-pitted top layer **105** are enlarged. Advantageously, such enlargement provides room for the second epitaxial crystalline layer **108** of undoped InGaN to expand and thus strain relax, as well as providing a route for electrical connection to the n-type crystalline template layer **104** through the epitaxial structure **300B**. Advantageously, forming V-grooves **302** in this manner provides a method for tuning the width of the gap in which material is subsequently formed. This is beneficial where, for example, lithographic techniques for forming V-grooves in an already formed epitaxial structure, in order to provide conduction paths to the n-type material, are not practical for forming V-grooves on a sufficiently small scale.

Strain relaxation is facilitated by annealing the structure **300B** of FIG. **3B**. In further examples, strain relaxation is facilitated using additional and/or alternative techniques. Once the sidewalls **305**, **306**, **308** of the V-groove **302** have been etched, the epitaxial structure **300B** is processed further, as shown at FIG. **3D**. FIG. **3C** shows a plan view **300C** of the structure of FIG. **3B**. There is shown randomly distributed V-pits **202** in the surface of the second epitaxial crystalline layer **108**. There is also shown a V-groove **302** forming a hexagonal pattern. Accordingly, the V-groove **302** isolates portions of the second epitaxial crystalline layer **108** of InGaN in order to form islands or domains by valleys at least partially forming the perimeter of one or more islands in the second epitaxial crystalline layer **108**, such that the valleys at least partially surround one or more islands in the second epitaxial crystalline layer **108**. Whilst the V-groove **302** is shown in a hexagonal configuration, in further examples, the V-groove **302** is alternatively and/or additionally formed in any appropriate configuration. Advantageously, the formation of two dimensional islands in the second epitaxial crystalline layer **108** based on such litho-

graphic patterning and etching of the crystalline template layer **104** provides strain relaxed islands with lateral dimensions of the order of 1 μm to 10 μm . Such formation of islands is highly controllable and provides relaxed island domains comparable with the size of devices that are formable on the relaxed islands domains, such as micro LED devices with a pixel size of the order of 1 μm to 10 μm . In further examples, the lateral dimensions of the islands are controlled to provide strain relaxed islands of dimensions appropriate to the formation of device based on subsequently formed crystalline layers.

Once the epitaxial structure **300B** of FIG. **3B** has been provided, the epitaxial structure **300B** is then planarized. This is shown at FIG. **3D**. FIG. **3D** shows that is the structure **300D** that is the epitaxial structure **300B** of FIG. **3B** that has been processed further. There is shown n-type material **210** formed in the channels provided by the V-grooves **302** and the V-pits **202**. The random V-pits **202** filled with the n-type material **210** provide uniform electron injection into subsequent layers. As with the epitaxial structure **200C** of FIG. **2C**, a subsequent pseudomorphic layer **212** is formed upon the planarized layer of n-type material **210**. The planarized layer of n-type material **210** is under tensile stress when formed from n-GaN, as the second epitaxial crystalline layer **108** beneath it is relaxed. Whilst the conductive material formed in the one or more channels provided by the V-grooves **302** and/or etched V-grooves **302** is a conductive n-type material **210** that is shown to planarize the structure, in further examples, alternatively or additionally, the conductive n-type material **210** partially planarizes the second epitaxial crystalline layer **108** or does not planarize the second epitaxial crystalline layer **108**, whilst still at least partially filling the one or more channels provided by V-pits **202** and/or V-grooves **302** with conductive material thereby to provide conductivity to relatively strain relaxed regions formed subsequently to the relaxation of the second epitaxial crystalline layer **108**.

Whilst the further layer **212** is shown to be provided on the n-type material **210**, in further examples the further layer **212** is provided on both the n-type material **210** and the second epitaxial crystalline layer **108**. In further examples, the further layer **212** comprises one or more LED devices, which, in a further example, form an array of devices, such as a high resolution array of micro LED devices. In further examples, the further layer **212** alternatively and/or additionally forms part of the n-type material **210** and/or replaces the n-type material, thereby beneficially to enable carrier injection directly into the portion of the strain relaxed material formed on the second epitaxial crystalline layer **108** through the further layer **212** formed at least partially on a sidewall of the n-type crystalline template layer **104**, such as the sidewall of the enlarged V-grooves **302**.

Whilst photolithography and etching techniques are used to form the trenches **304** in the crystalline template layer **104**, in further examples different patterning techniques are used. Further, whilst the relative cross sectional size of the V-pits **202** and the V-grooves **302** are shown in FIGS. **3A** to **3D** such that the V-grooves **302** are significantly deeper than the V-pits **202**, in further examples, the V-grooves **302** and V-pits **202** have different relative sizes. The V-pits **202** and the V-grooves **302** advantageously effectively provide channels for the electrical connection of the crystalline template layer **104** with layers formed on the insulating second epitaxial crystalline layer **108**. Beneficially, V-pits **202** aid uniform electron injection into subsequently grown layers.

Advantageously, V-grooves **302** aid relaxation of the second epitaxial crystalline layer **108** and V-pits **202** provide

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channels for electrical connection with the crystalline template layer **104**. If the subsequently formed devices, such as LED devices providing pixels with defined light emitting surfaces have a light emitting surface area greater than the V-groove **302** density, the structure can be provided without V-pits **202**, as sufficient current spreading may be enabled through the V-grooves **302**. If the light surface area of a pixel, for example, is less than the density of V-grooves **302**, V-pits **202** are particularly beneficial in providing sufficient current spreading in the LED device associated with the pixel.

FIG. **4A** shows a cross section view of an epitaxial structure **400A** that is a processed version of the epitaxial structure **100B** of FIG. **1B**. In order to enable lateral relaxation of the second epitaxial crystalline layer **108** following porosification of the first epitaxial crystalline layer **106** to provide the porosified first epitaxial crystalline layer **106'**, channels are formed through the epitaxial structure **400A**.

The epitaxial structure **400A** is patterned on the surface of the second epitaxial crystalline layer **108** using a self-assembled hard mask. A thin film of nickel (Ni) is evaporated on the surface of the second epitaxial crystalline layer **108**. The thin film of nickel is subsequently annealed to form random droplets. A dry etch is then used to form channels **402** through the epitaxial structure **400A** in order to provide channels to the conductive n-type crystalline template layer **104**. The nickel mask is then removed using a wet clean technique. Whilst porosification of the first epitaxial crystalline layer **106** occurs prior to the channels **402** being formed, in further examples, alternatively or additionally, channels **402** are formed before the porosification of the first epitaxial crystalline layer **106**. Advantageously, the channels **402** formed through the first epitaxial crystalline layer **106** (or porosified first epitaxial crystalline layer **106'**) and the second epitaxial crystalline layer **108** at least partially define the perimeter of two dimensional domains or islands in the second epitaxial crystalline layer **108**.

Once the channels **402** have been formed, the process moves to FIG. **4B**. At FIG. **4B** there is shown a cross sectional view of an epitaxial structure **400B** that is the epitaxial structure **400A** of FIG. **4A** that has been processed further in order to form a planarized layer **404** of material in the channels **402**. Whilst the conductive material **404** formed in the one or more channels is an n-type material **404**, such as n-type GaN, that is shown to planarize the structure, in further examples, alternatively or additionally, the conductive n-type material **404** partially planarizes the second epitaxial crystalline layer **108** or does not planarize the second epitaxial crystalline layer **108**, whilst still at least partially filling the one or more channels provided by channels **402** with conductive material **404** thereby to provide conductivity to relatively strain relaxed regions formed subsequently to the relaxation of the second epitaxial crystalline layer **108**.

A further layer **412**, which is an example InGaN red light emitting active region is then formed on the material **404** in an analogous fashion to the further layer **212** described with reference to FIGS. **2** and **3**. Such a further layer **412**, in further example, forms one or more LED structures. In further examples, such LED structures form part of an array of LED devices, such as a high resolution array of micro LED devices. Whilst the further layer **412** is shown to be provided on the n-type material **404**, in further examples the further layer **412** is provided on both the n-type material **404** and the second epitaxial crystalline layer **108**. In further examples, the further layer **412** comprises one or more LED devices, which, in a further example, form an array of

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devices, such as a high resolution array of micro LED devices. In further examples, the further layer **412** alternatively and/or additionally forms part of the n-type material **404** and/or replaces the n-type material, thereby beneficially to enable carrier injection directly into the portion of the strain relaxed material formed on the second epitaxial crystalline layer **108** through the further layer **412** formed at least partially on a sidewall of the n-type crystalline template layer **104**, such as a sidewall of the channel **402** formed through to the crystalline template layer **104**.

Advantageously, self-assembled nickel droplets are small and dense, very narrow gaps can be created such that strain relaxation is easier, as well as planarization. Planarization can be achieved over a short distance for narrow gaps. Further filling such a dense distribution of gaps with the n-type material **210** provides good current flow through the epitaxial structure **400B**.

Beneficially, the use of a hard mask, such as a self-assembled nickel hard mask described with respect to FIGS. **4A** and **4B** enables the formation of densely packed two dimensional islands in the second epitaxial crystalline layer **108** with lateral dimensions of the order of between 250 nm and 1000 nm. Advantageously, islands formed on such a scale not only facilitate planarization, but they also provide for a uniform strain relaxed layer where the lateral dimensions of the islands are less than the lateral dimensions of the devices formed in layer upon the relaxed islands, such as micro LED devices where the pixel size is typically greater than 1000 nm. This means that, effectively, a size-agnostic strain relaxation layer can be provided where the scale of the relaxed islands is less than the scale of the functioning devices formed on the relaxed islands.

At FIGS. **5A** and **5B** there is shown the formation of the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108** on a crystalline template layer **104** with a morphology provided by patterning the crystalline template layer **104**.

FIG. **5A** shows a cross sectional view of an epitaxial structure **500A** formed on a patterned substrate. There is shown a crystalline template layer **104** as described with reference to FIGS. **1** to **4**. The crystalline template layer **104** is patterned using known techniques to deposit a mask **502**. Whilst the mask **502** is shown in cross section, the skilled person understands that the mask **502** is formable in two dimensions to provide an array of apertures on the crystalline template layer. The mask **502** is formed using silicon dioxide. In further examples, the mask **502** is alternatively and/or additionally formed using further materials. Overgrowth **504** of n-type GaN on the masked crystalline template layer results in the overgrowth **504** forming a structures of GaN in electrical communication with the n-type GaN crystalline template layer **104**. The overgrowth **504** is shown to form trapezoidal cross section structures in FIG. **5A**, however, the skilled person understands that the morphology of such overgrowth **504** extends to 3 dimensions and is related to the underlying crystalline structure of the material used for overgrowth **504** structures. Once grown, the overgrowth **504** structures form part of the crystalline template layer **104**.

The dimensions of the trapezoidal overgrowth **504** structures described with respect to FIGS. **5A** and **5B** vary as a function of the trapezoidal pyramid height. The lateral dimension of the trapezoidal features is of the order of 4 μm at the base of the trapezoidal feature and of the order of 3 μm at the top of the trapezoidal feature, for a height of approximately 1 μm . In further examples, the lateral dimensions of the trapezoidal features are controlled to provide alternative

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or additional features of different sizes. In further examples, the dimensions of the overgrowth **504** features are controlled to provide laterally sized islands in the second epitaxial crystalline layer **108** suitable for the formation of microLED devices with pixel dimensions of the order of between 1 μm and 10 μm . Alternatively, or additionally, island domains in the second epitaxial crystalline layer **108** with lateral dimensions less than 1 μm are provided. Advantageously, the formation of two dimensional islands in the second epitaxial crystalline layer **108** based on such lithographic patterning and overgrowth provides strain relaxed islands with lateral dimensions of the order of 1 μm to 10 μm , or less. Such formation of islands is highly controllable and provides relaxed island domains comparable with the size of devices that are formable on the relaxed islands domains, such as micro LED devices with a pixel size of the order of 1 μm to 10 μm .

Once the overgrowth **504** structures have been formed, an initially highly doped first epitaxial crystalline layer **106**, such as that described with reference to FIGS. **1** to **4**, is formed. The first epitaxial crystalline layer **106** is formed on the top and the sidewalls of the overgrowth **504** formed in the apertures of the mask **502**. The thickness of the first epitaxial crystalline layer **106** on the top of the overgrowth **504** features is approximately 30 nm. The growth rates of crystalline materials on different crystalline planes, such as those provided by the top and sidewalls of the overgrowth **504** structures may be different and can be controlled by altering the growth rate of crystalline material, as well as the composition of the crystalline material, for example. Therefore, the thickness of crystalline layer deposited on the different faces of the overgrowth **504** structures may be different. The thickness of the first epitaxial crystalline layer **106** and the second epitaxial crystalline layer **108** are described above with respect to the thickness of layers generally perpendicular to the underlying planar substrate (and therefore parallel to the primary growth direction). Where the thickness of the first epitaxial crystalline layer **106** on the top of the overgrowth **504** features is approximately 30 nm, the thickness of the first epitaxial crystalline layer **106** on the sidewalls of the overgrowth **504** features is of the order of 1 to 3 nm. In further examples, the thickness of the first epitaxial crystalline layer **106** on the sidewalls is a different thickness. Beneficially, the relatively thinner first epitaxial crystalline layer **106** on the sidewalls of the overgrowth **504** features means that the first epitaxial crystalline layer **106** on the sidewalls of the overgrowth **504** features can be etched off straightforwardly, as described below, in order to provide a channel to the underlying n-GaN.

Subsequently, a second epitaxial crystalline layer **108**, such as that described with reference to FIGS. **1** to **4** is formed on the first epitaxial crystalline layer **106**. As described above with reference to FIGS. **1** to **4**, the second epitaxial crystalline layer **108** is initially strained by the underlying material of the crystalline template layer **104** and the first epitaxial crystalline layer **106** such that the different inherent, natural, in-plane lattice parameter of the second epitaxial crystalline layer **108** is pseudomorphically aligned with the inherent, natural, in-plane lattice parameter of the first epitaxial crystalline layer **106**. The second epitaxial crystalline layer **108** has a thickness of between 100 nm and 150 nm on top of the overgrowth **504** features. As described above with respect to the first epitaxial crystalline layer **106** in FIG. **5A**, the growth rates of crystalline materials on different crystalline planes, such as those provided by the top and sidewalls of the overgrowth **504** structures may be different. Where the thickness of the second epitaxial crys-

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talline layer **108** on the top of the overgrowth **504** features is approximately 100 to 150 nm, the thickness of the second epitaxial crystalline layer **108** on the sidewalls of the overgrowth **504** features is of the order of 5 to 20 nm. In further examples, the thickness of the second epitaxial crystalline layer **108** on the sidewalls is a different thickness.

Electrochemical etching of the epitaxial structure **500A** results in porosification of the first epitaxial crystalline layer **106** through threading dislocations or other openings in the undoped InGaN of the second epitaxial crystalline layer **108** enabling such electrochemical processing of the subsurface first epitaxial crystalline layer **106** to take place. This is facilitated by the high doping contrast between the n-type crystalline template layer **104** and first epitaxial crystalline layer **106** meaning that only the first epitaxial crystalline layer **106** becomes porosified during an electrochemical etching of this layer. Advantageously, the doping contrast enables control of the porosification process since, as the first epitaxial crystalline layer **106** has pores formed in it, the porosified first epitaxial crystalline layer **106'** becomes highly resistive and therefore the electrochemical process used to porosify the first epitaxial crystalline layer **106** stops forming pores. The porosified first epitaxial crystalline layer **106'** is shown at FIG. **5B**.

FIG. **5B** shows an epitaxial structure **500B** that is the structure **500A** of FIG. **5A** that has been processed further to provide an array of light emitting structures. There is shown the porosified first epitaxial crystalline layer **106'** formed on the top of the overgrowth **504** structures and the second epitaxial crystalline layer **108** formed on the porosified first epitaxial crystalline layer **106'**. The porosification of the first epitaxial crystalline layer **106** results in the removal of material in the channels between overgrowth **504** structures from the sidewalls of the overgrowth **504** structures. In further examples, material from the sidewalls of the overgrowth **504** structures is removed by different means to form channels through the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108** to form a path through layers **106'**, **108** to the n-type GaN of the overgrowth **504** structures of GaN formed on the crystalline template layer **104**. Beneficially, in GaN based structures, such as the epitaxial structures **500A**, **500B** described with reference to FIGS. **5A** and **5B**, the growth direction is typically perpendicular to the c-plane and the etching rate of inclined facets, such as the sidewalls of the overgrowth **504** structures is faster than that of the c-plane material, such as that formed on the top of the overgrowth **504** structures. This means that the porosified first epitaxial crystalline layer **106'** and the second first epitaxial crystalline layer **108** can be etched from the sidewalls of the overgrowth **504** structures at a faster rate than the material formed on top of the overgrowth **504** structures is etched. Accordingly, removal of material to form channels to the n-GaN through the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108** is facilitated.

Removal of the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108** effectively creates 2 dimensional domains or islands in the second epitaxial crystalline layer **108**. The porosification of the first epitaxial crystalline layer **106** enables strain relaxation in the second epitaxial crystalline layer **108** of undoped InGaN. Such strain relaxation requires the second epitaxial crystalline layer **108** to expand. The lateral expansion of the second epitaxial crystalline layer **108** is controlled in order to provide advantageous structures, as described herein. The extent to which the porosified first epitaxial crystalline layer **106'** is, by density of volume, pore and GaN material,

depends on the electrochemical etching process. The electrochemical etching process, whilst beneficially providing compliant material in a subsurface layer, results in the porosified first epitaxial crystalline layer **106'** having a reduced electrical conductivity compared with the highly doped conductivity of the first epitaxial crystalline layer **106** when initially formed. Beneficially, removal of the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108** effectively to create 2 dimensional domains or islands in the second epitaxial crystalline layer **108** also creates channels for the electrical connection of subsequently grown layers on grown on the second epitaxial crystalline layer **108** with the crystalline template layer **104**.

A further layer **506**, analogous to the further layers **212**, **414** described above with respect to FIGS. **2** to **4** is conformally deposited and formed on the relaxed second epitaxial crystalline layer **108**. The further layer **506** is an active region. The further layer **506** includes one or more quantum wells, such as InGa_N based quantum wells designed to emit light with a primary wavelength of light corresponding to red light upon carrier injection and recombination. In further examples, the further layer **506** comprises additional and/or alternative layers that benefit from formation on the relatively relaxed second epitaxial crystalline layer **108**. The further layer **506** is a third epitaxial crystalline layer that provides a channel for electrical conductivity from the n-GaN overgrowth **504** forming part of the crystalline template layer **104** to the portion of the active region formed on the relaxed second epitaxial crystalline layer **108**.

Upon the further layer **506** there is formed a p-type region **508**. The p-type region **508** is conformally deposited and formed on the further layer **506** on the top of the overgrowth **504** structures and on the sidewalls of the overgrowth **504** structures. The p-type region **508** is formed from p-type doped GaN. In further examples, additional and/or alternative layers are used to form the p-type region **508**.

Advantageously, the structure **500B** of FIG. **5B** provides an array of light emitting diode structures with active regions formed on relatively strain relaxed layers with lattice constants closer to the lattice constant of the material forming the active region, thereby to provide high quality light emitting regions with improved efficiency. The formation of such active regions on overgrowth **504** structures improves light extraction from devices based on such high quality light emitting regions and improves collimation.

Beneficially, where the third epitaxial layer forming electrical connection through the porosified first epitaxial crystalline layer **106'** and the second epitaxial crystalline layer **108** forms part of an active region on one or more sidewalls of one or more of the overgrowth **504** structures, a direct current path for electrons between the crystalline template layer **104** and the active region of the further layer **506** on the island of the second epitaxial crystalline layer **108** is provided. Advantageously, such injection is controlled and sidewall injection enables a reduced forward voltage for multiple quantum well (MQW) structures forming part of the further layer **506** where there is a high operating current.

Whilst FIG. **5B** shows the mask **502**, in further examples the mask **502** is removed after the overgrowth **504** step. Further, whilst the further layer **506** and the p-type region **508** are conformally formed on the overgrowth **504** structures, individual structures are optionally disconnected thereby to enable the formation of multiple, individually addressable light emitting diode structures with a common n-type electrode formed by the crystalline template layer **104**.

Advantageously, the techniques and structures described herein provide a way of forming a relatively strain relaxed layer upon a conductive template layer whilst enabling electrical connection between the conductive template layer and layers subsequently grown on the relatively strain relaxed layer. Advantageously, where an epitaxial crystalline layer with an inherent in-plane lattice parameter that is different to the inherent in-plane lattice parameter of a crystalline template layer is formed on an intervening layer such that the epitaxial crystalline layer is pseudomorphically formed with respect to the intervening layer, porosification of the intervening layer in conjunction with the formation of channels through the porosified intervening layer and the epitaxial crystalline layer enables strain relaxation in the epitaxial crystalline layer as well as a route for current injection in devices formed on the relaxed epitaxial crystalline layer.

Beneficially, channels formed and filled with material to enable electrical connection overcome difficulties associated with the use of highly resistive layers used for strain relaxation and hence the provision of functioning devices. Advantageously, the use of channels with faces angled with respect to the direction perpendicular to the planar growth direction of epitaxial layers provides for sloped surfaces upon which connection can be made to underlying crystalline template layers whilst simultaneously providing strain relaxed light emitting structures on disconnected islands of relatively strain relaxed material that has a different inherent in-plane lattice parameter to the inherent in-plane lattice parameter of the crystalline template layer above which the relatively strain relaxed material is formed.

Beneficially, light emitting diode (LED) devices can be formed on the structures described with reference to FIGS. **1** to **5**. Advantageously, arrays of LEDs with associated pixels can be formed on the strain relaxed layers whilst enabling n-type contact through the n-type region even though the strain relaxation uses a porous layer that is highly resistive. This avoids the need to form conductive layers of material and the formation of contacts for both p-type and n-type layers in LED devices from the same side of the device. This is particularly advantageous, for example, in respect of longer wavelength InGa_N based light emitting devices, such as LEDs formed from nitride material configured to emit light with a primary peak wavelength that corresponds to red light, where a strain relaxed islands of undoped InGa_N can be used instead of thicker, conducting layers of InGa_N which are typically of poor quality. Further, enabling conduction to make n-type contacts on the opposite side of the device from the p-type contact means that pixels associated with LED devices in arrays can be packed more closely, thereby improving the resolution of display devices based on such arrays.

Whilst the processes described herein are given in a particular order, the skilled person understands that in further examples, alternative and/or additional steps and process orders are implemented whilst enabling the functionality provided.

The invention claimed is:

1. A method of forming a strain relaxation layer in an epitaxial crystalline structure, the method comprising:
 - providing a crystalline template layer comprising a material with a first natural relaxed in-plane lattice parameter;
 - forming a first epitaxial crystalline layer on the crystalline template layer, wherein the first epitaxial crystalline

layer has an initial electrical conductivity that is higher than the electrical conductivity of the crystalline template layer;

forming a second epitaxial crystalline layer on the first epitaxial crystalline layer, wherein the second epitaxial crystalline layer has an electrical conductivity lower than the initial electrical conductivity of the first epitaxial crystalline layer and comprises a material with a second natural relaxed in-plane lattice parameter that is different to the first natural relaxed in-plane lattice parameter of the crystalline template layer;

forming pores in the first epitaxial crystalline layer by electrochemical etching of the first epitaxial crystalline layer to enable strain relaxation in the second epitaxial crystalline layer by plastic deformation of bonds in the first epitaxial crystalline layer and/or at the interface between the first epitaxial crystalline layer and the second epitaxial crystalline layer; and

forming one or more channels comprising a conductive material through at least the first epitaxial crystalline layer and the second epitaxial crystalline layer thereby to enable electrical connection to the crystalline template layer through the first epitaxial crystalline layer and the second epitaxial crystalline layer.

2. The method according to claim 1, wherein the pores are formed with a density of the volume of the first epitaxial layer between the crystalline template layer and the second epitaxial crystalline layer of greater than 50%, preferably greater than 60% and more preferably greater than 70%.

3. The method according to claim 1, wherein the second epitaxial crystalline layer comprises at least one V-pit, preferably wherein the depth of the at least one V-pit is substantially the thickness of the first epitaxial crystalline layer and the second epitaxial crystalline layer combined.

4. The method according to claim 3, comprising enlarging the at least one V-pit, preferably wherein enlarging the at least one V-pit comprises etching material from at least one sidewall of the at least one V-pit thereby to expose at least a portion of the crystalline template layer.

5. The method according to claim 1, comprising etching through at least the second epitaxial crystalline layer thereby to form one or more islands in the second epitaxial crystalline layer, preferably comprising patterning the second epitaxial crystalline layer, wherein patterning comprises at least one of a lithographic technique and a self-assembled Ni hard mask.

6. The method according to claim 1, comprising forming the crystalline template layer on a patterned template layer.

7. The method according to claim 6, wherein the patterned template layer is patterned to provide trenches, preferably wherein forming the crystalline template layer comprises forming overgrowth material in the trenches to provide V-grooves, thereby to enable the formation of two dimensional domains at least partially defined by valleys in the second epitaxial crystalline layer.

8. The method according to claim 6, wherein the crystalline template layer is patterned to provide one or more overgrowth structures, preferably wherein forming the patterned template layer comprises at least partially masking the crystalline template layer to provide one or more apertures and forming overgrowth material in the one or more apertures thereby to provide the one or more overgrowth structures, preferably wherein forming the one or more channels comprising the conductive material through at least the first epitaxial crystalline layer and the second epitaxial crystalline layer comprises removing material from one or more sidewalls of the one or more overgrowth structures.

9. The method according to claim 1, comprising planarizing at least a portion of the second epitaxial crystalline layer thereby to form a third epitaxial crystalline layer that is pseudomorphic with the second epitaxial crystalline layer, preferably wherein the conductive material forms part of the third epitaxial crystalline layer.

10. The method according to claim 9, wherein the third epitaxial crystalline layer is an electrically conductive layer, preferably wherein the third epitaxial crystalline layer is in electrical communication with the crystalline template layer and/or wherein the third epitaxial crystalline layer forms at least part of an active region configured to emit light in response to carrier recombination.

11. The method of claim 9, wherein at least one of the crystalline template layer, the first epitaxial crystalline layer, the second epitaxial crystalline layer and the third epitaxial crystalline layer comprises a semiconductor material, preferably wherein the semiconductor material is a III-V based material, more preferably wherein the III-V based material is a nitride based material.

12. An epitaxial crystalline structure comprising:

a crystalline template layer comprising a material with a first natural relaxed in-plane lattice parameter;

a first epitaxial crystalline layer formed on the crystalline template layer;

a second epitaxial crystalline layer formed on the first epitaxial crystalline layer, wherein the second epitaxial crystalline layer comprises a material with a second natural relaxed in-plane lattice parameter that is different to the first natural relaxed in-plane lattice parameter of the crystalline template layer;

one or more pores formed in the first epitaxial crystalline layer such that any strain in the second epitaxial crystalline layer is relaxed by plastic deformation of bonds in the first epitaxial crystalline layer and/or at the interface between the first epitaxial crystalline layer and the second epitaxial crystalline layer; and

one or more channels comprising a conductive material formed through at least the first epitaxial crystalline layer and the second epitaxial crystalline layer thereby to enable electrical connection to the crystalline template layer through the first epitaxial crystalline layer and the second epitaxial crystalline layer.

13. The epitaxial crystalline structure according to claim 12, comprising a third epitaxial crystalline layer that is pseudomorphic with the second epitaxial crystalline layer.

14. The epitaxial crystalline structure according to claim 13, wherein the conductive material forms part of the third epitaxial crystalline layer.

15. The epitaxial crystalline structure according to claim 12, wherein the one or more pores are formed with a density of the volume of the first epitaxial layer between the crystalline template layer and the second epitaxial crystalline layer of greater than 50%, preferably greater than 60% and more preferably greater than 70%.

16. The epitaxial crystalline structure according to claim 12, wherein the second epitaxial crystalline layer comprises at least one V-pit, preferably wherein the depth of the at least one V-pit is substantially the thickness of the first epitaxial crystalline layer and the second epitaxial crystalline layer combined.

17. The epitaxial crystalline structure according to claim 16, wherein the at least one V-pit exposes at least a portion of the crystalline template layer.

18. The epitaxial crystalline structure according to claim 12, comprising one or more islands formed in the second epitaxial crystalline layer, and comprising one or more

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valleys at least partially surrounding the one or more islands in the second epitaxial crystalline layer thereby to enable the formation of one or more two dimensional domains.

19. The epitaxial crystalline structure according to claim **12**, wherein the third

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epitaxial crystalline layer is an electrically conductive layer, preferably wherein the third epitaxial crystalline layer is in electrical communication with the crystalline template layer and/or wherein the third epitaxial crystalline layer forms part of an active region configured

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to emit light in response to carrier recombination.

20. A light emitting device comprising the epitaxial crystalline structure according to claim **12**.

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